

2016

Theory of Computing

Paper : MCA-303

Full Marks : 70

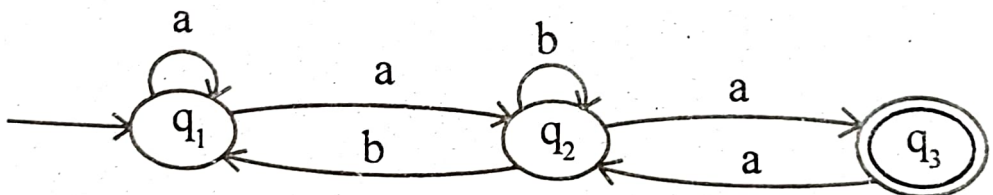
Time : 3 Hours

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

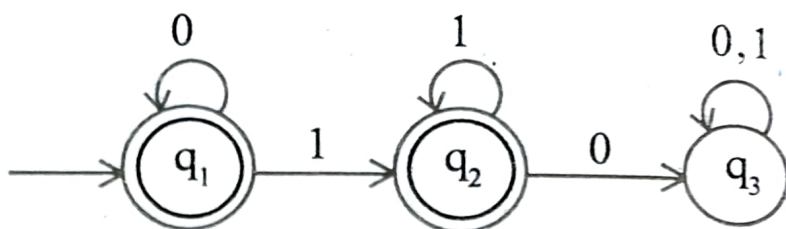
Answer any five questions.

1. a) Let $G = (\{G, C\}, \{a, b\}, P, S)$ where P consists of $S \rightarrow aCa$, $C \rightarrow aCa \mid b$. Find $L(G)$.
- b) If G is $S \rightarrow aS \mid bs \mid a \mid b$, find $L(G)$. 7+7=14
2. a) Consider the transition system given in figures prove that the strings recognized are $(a+ a(b+aa)^*b)^*(b+aa)^*a$.



- b) Describe in English the aspect by FA whose transition diagram is given in figure.

[Turn over]



$$7+7=14$$

3. a) Construct a DFA for regular expression $r = ba + (a + bb)a^*b$.

b) Construct a transition graph for regular expression $R = (0(011)^*)^+$. $7+7=14$

4. a) Consider G whose productions are $S \rightarrow aAS \mid a$, $A \rightarrow SbA \mid SS \mid ba$. Show that $S \rightarrow aabbbaa$ and construct a derivation tree whose yield is $aabbbaa$.

b) If G is the grammar $S \rightarrow SbS \mid a$. Show that G is ambiguous. $7+7=14$

5. a) Consider a grammar $G = (\{S, A, B\}, \{a, b\}, P, S)$ where S is the start symbol and P is given by $S \rightarrow bA \mid aB$, $A \rightarrow bAA \mid aS \mid a$, $B \rightarrow aBB \mid bs \mid b$. Convert it an equivalent Chomsky normal form.

b) Convert the following grammar into CNF. $S \rightarrow aXX$, $X \rightarrow aS \mid bS \mid a$. $7+7=14$

6. Design a PDA for the language $L = \{a^n b^n | n > 0\}$.

14

7. For the grammar $G = (\{S, A, B\}, \{a, b\}, P, S)$ where P consists of $S \rightarrow aAB$, $S \rightarrow bBA$, $A \rightarrow bS$, $A \rightarrow a$, $B \rightarrow as$, $B \rightarrow b$, $W = abbbab$ is in $L(G)$. Construct a deterministic PDA accepting $L(G)$ and a leftmost derivation of $abbbab$.

14